Supplementary Material for “Do Selected Neural and Zero-Shot Time-Series Models Beat Persistence in Absolute Stock-Price Forecasting? A Cross-Market Audit”

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1 Purpose

This supplement reports expanded CSV-derived tables for the cross-market baseline-first stock-price forecasting audit. The main paper keeps only compact evidence tables. This document preserves additional model, dataset, and row-level summaries so that the negative conclusion can be inspected without overloading the main manuscript.

2 Included Artifacts

The evidence package contains the following generated files:

- `combined_model_results.csv`: unified supervised and Chronos result rows.
- `model_gate_summary.csv`: model-level pass rates and RMSE-ratio confidence intervals.
- `model_horizon_summary.csv`: horizon-level summaries.
- `dataset_model_summary.csv`: dataset-by-model summaries.
- `best_100_model_rows.csv` and `worst_100_model_rows.csv`: row-level extremes.
- `chronos_full_origin_subset_60_rows.csv`: the 60-row full-origin Chronos subset verification cited in the main manuscript.
- `FULL_ORIGIN_SUBSET_SUMMARY.md`: a compact summary of the full-origin subset verification.

Table 1. Protocol-level model summary.

Protocol	Model	Rows	Passes	Pass rate	Median ratio
full rolling-origin supervised audit	DLinear	5940	0	0.00%	3.150
full rolling-origin supervised audit	GRU	5940	5	0.08%	2.221
full rolling-origin supervised audit	LSTM	5940	8	0.13%	2.641
full rolling-origin supervised audit	PatchTST-style	5940	3	0.05%	3.766
full rolling-origin supervised audit	TimeMixer-style	5940	3	0.05%	2.328
full rolling-origin supervised audit	iTransformer-style	5940	4	0.07%	3.723
sampled-origin zero-shot foundation audit	Chronos-T5-small zero-shot	990	133	13.43%	1.035

Table 2. Dataset-by-model persistence-gate summary.

Dataset	Model	Rows	Passes	Pass rate	Median ratio
Dow 30	Chronos-T5-small zero-shot	180	21	11.67%	1.036
Dow 30	DLinear	1080	0	0.00%	2.671
Dow 30	GRU	1080	1	0.09%	2.390
Dow 30	LSTM	1080	2	0.19%	2.794
Dow 30	PatchTST-style	1080	0	0.00%	4.078
Dow 30	TimeMixer-style	1080	0	0.00%	2.324
Dow 30	iTransformer-style	1080	0	0.00%	4.161
ETF basket	Chronos-T5-small zero-shot	72	13	18.06%	1.032
ETF basket	DLinear	432	0	0.00%	3.676
ETF basket	GRU	432	1	0.23%	4.364
ETF basket	LSTM	432	0	0.00%	5.233
ETF basket	PatchTST-style	432	0	0.00%	5.553
ETF basket	TimeMixer-style	432	0	0.00%	3.039
ETF basket	iTransformer-style	432	0	0.00%	6.224
Hang Seng large-cap	Chronos-T5-small zero-shot	258	40	15.50%	1.033
Hang Seng large-cap	DLinear	1548	0	0.00%	4.747
Hang Seng large-cap	GRU	1548	2	0.13%	1.408
Hang Seng large-cap	LSTM	1548	4	0.26%	1.639
Hang Seng large-cap	PatchTST-style	1548	3	0.19%	2.085
Hang Seng large-cap	TimeMixer-style	1548	3	0.19%	1.970
Hang Seng large-cap	iTransformer-style	1548	4	0.26%	1.959
Nasdaq large-tech	Chronos-T5-small zero-shot	180	19	10.56%	1.035
Nasdaq large-tech	DLinear	1080	0	0.00%	2.291
Nasdaq large-tech	GRU	1080	0	0.00%	3.996
Nasdaq large-tech	LSTM	1080	0	0.00%	4.572
Nasdaq large-tech	PatchTST-style	1080	0	0.00%	5.108
Nasdaq large-tech	TimeMixer-style	1080	0	0.00%	2.404
Nasdaq large-tech	iTransformer-style	1080	0	0.00%	5.364
S&P 100 subset	Chronos-T5-small zero-shot	300	40	13.33%	1.036
S&P 100 subset	DLinear	1800	0	0.00%	3.000
S&P 100 subset	GRU	1800	1	0.06%	2.412
S&P 100 subset	LSTM	1800	2	0.11%	2.868
S&P 100 subset	PatchTST-style	1800	0	0.00%	4.189
S&P 100 subset	TimeMixer-style	1800	0	0.00%	2.359
S&P 100 subset	iTransformer-style	1800	0	0.00%	4.229

Best model rows by RMSE ratio.

Dataset	Ticker	Model	Horizon	Price	Ratio	Pass
ETF basket	SLV	Chronos-T5-small zero-shot	10	raw_close	0.916	1
S&P 100 subset	BK	Chronos-T5-small zero-shot	10	raw_close	0.922	1
Nasdaq large-tech	AMD	Chronos-T5-small zero-shot	1	raw_close	0.930	1
Hang Seng large-cap	0267.HK	PatchTST-style	10	raw_close	0.939	1
Dow 30	KO	Chronos-T5-small zero-shot	5	adjusted_close	0.942	1
Hang Seng large-cap	0267.HK	Chronos-T5-small zero-shot	1	adjusted_close	0.947	1
Hang Seng large-cap	0941.HK	Chronos-T5-small zero-shot	5	raw_close	0.947	1
Hang Seng large-cap	3690.HK	Chronos-T5-small zero-shot	10	raw_close	0.952	1
S&P 100 subset	GS	Chronos-T5-small zero-shot	10	adjusted_close	0.955	1
Hang Seng large-cap	1109.HK	Chronos-T5-small zero-shot	10	adjusted_close	0.956	1
Hang Seng large-cap	0267.HK	Chronos-T5-small zero-shot	1	raw_close	0.956	1
ETF basket	GLD	Chronos-T5-small zero-shot	10	adjusted_close	0.957	1
Hang Seng large-cap	0992.HK	Chronos-T5-small zero-shot	10	raw_close	0.958	1
ETF basket	SLV	Chronos-T5-small zero-shot	5	raw_close	0.961	1
S&P 100 subset	GOOG	Chronos-T5-small zero-shot	5	raw_close	0.961	1
Dow 30	TRV	Chronos-T5-small zero-shot	5	adjusted_close	0.963	1
S&P 100 subset	KO	Chronos-T5-small zero-shot	1	adjusted_close	0.965	1
Dow 30	CSCO	Chronos-T5-small zero-shot	5	raw_close	0.965	1
Hang Seng large-cap	0005.HK	Chronos-T5-small zero-shot	10	adjusted_close	0.965	1
S&P 100 subset	IBM	Chronos-T5-small zero-shot	5	raw_close	0.967	1

Worst model rows by RMSE ratio.

Dataset	Ticker	Model	Horizon	Price	Ratio	Pass
Hang Seng large-cap	1044.HK	TimeMixer-style	1	raw_close	84.389	0
Hang Seng large-cap	1044.HK	TimeMixer-style	1	raw_close	74.009	0
Hang Seng large-cap	1876.HK	DLinear	1	raw_close	71.852	0
Hang Seng large-cap	1876.HK	DLinear	1	raw_close	69.144	0
Hang Seng large-cap	1044.HK	TimeMixer-style	1	raw_close	67.975	0
Hang Seng large-cap	1876.HK	DLinear	1	adjusted_close	62.649	0
Hang Seng large-cap	1876.HK	DLinear	1	adjusted_close	60.096	0
ETF basket	HYG	DLinear	1	adjusted_close	53.897	0
Hang Seng large-cap	0003.HK	DLinear	1	raw_close	52.222	0
Hang Seng large-cap	1044.HK	TimeMixer-style	1	adjusted_close	50.238	0
ETF basket	HYG	DLinear	1	adjusted_close	46.881	0
Hang Seng large-cap	1876.HK	DLinear	1	raw_close	46.477	0
Hang Seng large-cap	1876.HK	DLinear	1	adjusted_close	41.395	0
Hang Seng large-cap	1044.HK	TimeMixer-style	1	adjusted_close	41.256	0
Hang Seng large-cap	2269.HK	DLinear	1	raw_close	40.595	0
Hang Seng large-cap	2269.HK	DLinear	1	adjusted_close	40.595	0
Hang Seng large-cap	1044.HK	TimeMixer-style	1	adjusted_close	40.178	0
S&P 100 subset	BK	DLinear	1	raw_close	38.740	0
S&P 100 subset	BK	DLinear	1	raw_close	37.917	0
ETF basket	TLT	DLinear	1	adjusted_close	37.739	0

3 Split Boundaries

The main paper uses chronological train/validation/test splits. Table S5 reports nominal split dates computed from the raw downloaded records. Exact model-row sample counts are retained in the evidence package because technical indicators and horizon-specific target shifting slightly change usable row counts.

Table S5. Nominal chronological split boundaries by dataset.

Dataset	Assets	First date	Train/validation boundary	Validation/test boundary	Last date
Dow 30	30	2015-01-02	2023-02-24	2024-01-23	2026-04-30
ETF basket	12	2015-01-02	2023-02-24	2024-01-23	2026-04-30
Hang Seng large-cap	43	2015-01-02	2023-02-22	2024-01-19	2026-04-30
Nasdaq large-tech	30	2015-01-02	2023-02-24	2024-01-23	2026-04-30
S&P 100 subset	50	2015-01-02	2023-02-24	2024-01-23	2026-04-30

Exact split indices can vary slightly after indicator construction and horizon-specific target shifting. The CSV file `split_boundary_summary.csv` records the raw-download nominal split boundaries, while model-result CSV files record train, validation, and test sample counts for each row.

4 Interpretation

The long-form tables reinforce the main result. Supervised models fail across nearly all dataset and model partitions. Chronos-T5-small is more competitive than the supervised implementations, but its wins are local rather than stable across the complete audit. These results justify the paper's restrained claim: persistence-first evaluation should precede architecture-level conclusions in absolute stock-price level forecasting.